

Maximum Principle for Absolute Minimisers

Simone Carano and Nikos Katzourakis

March 15, 2026

Abstract

2020 MSC:

Key words and phrases: Vectorial Calculus of Variations in L^∞ ; higher order problems; Absolute Minimisers, Maximum Principle.

1 Introduction

For $n, N, k \in \mathbb{N}$, let $\Omega \subset \mathbb{R}^n$ be an open bounded set and $H : \Omega \times \mathbb{R}^N \times \mathbb{R}^{Nn} \times \dots \times \mathbb{R}^{Nn^k} \rightarrow \mathbb{R}$ be $\mathcal{L}(\Omega) \otimes \mathcal{B}(\mathbb{R}^\alpha)$ -measurable, where we have identified $\mathbb{R}^\alpha \cong \mathbb{R}^N \times \mathbb{R}^{Nn} \times \dots \times \mathbb{R}^{Nn^k}$ for $\alpha = \alpha(n, N, k) = N \frac{n^{k+1}-1}{n-1}$. Here $\mathcal{L}(\Omega)$ is the Lebesgue σ -algebra of Ω and $\mathcal{B}(\mathbb{R}^\alpha)$ is the Borel σ -algebra of $\mathbb{R}^\alpha$. Let $U \subseteq \Omega$ be an open subset and consider the supremal functional

$$E_\infty(u, U) := \operatorname{ess\,sup}_{x \in U} H(x, D^{[k]}u(x)), \quad (1.1)$$

where $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ and $D^{[k]}u := (u, Du, \dots, D^k u)$ is the jet of order k of u .

The goal herein is to prove a maximum principle property for the absolute minimisers of the functional E_∞ . We recall that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser of E_∞ if

$$E_\infty(u, U) \leq E_\infty(u + \Phi, U) \quad \forall \Phi \in W_0^{k,\infty}(U, \mathbb{R}^N) \quad \forall U \subseteq \Omega \text{ open.}$$

Under very mild assumptions on the supremand H , we prove that for an absolute minimiser the essential supremum in (1.1) is “attained at the boundary of U ”. In order to give a precise meaning to this expression, we refer to the supremum on ∂U of the essential limsup of $H(\cdot, D^{[k]}u)$, which is defined as

$$H(x, D^{[k]}u(x))^* := \lim_{\varepsilon \rightarrow 0} \operatorname{ess\,sup}_{B_\varepsilon(x) \cap U} H(\cdot, D^{[k]}u) \quad \forall x \in \partial U. \quad (1.2)$$

Therefore, we prove that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ satisfies the following

$$\operatorname{ess\,sup}_U H(\cdot, D^{[k]}u) = \sup_{\partial U} H(\cdot, D^{[k]}u)^* \quad \forall U \subseteq \Omega \text{ open.}$$

This maximum principle property is only necessary for absolute minimisers: the cone function $u(x) = |x|$ belongs to $W^{1,\infty}(\mathbb{R}^n)$ and clearly satisfies the maximum principle for $E_\infty(u, \Omega) := \operatorname{ess\,sup}_\Omega |Du|$, but it is not an absolute minimiser. However, we will see that the maximum principle is a property that characterises a class of maps which is larger than the one of absolute minimisers, and are the ones that minimise E_∞ with respect to compactly supported variations.

Before stating our main results, we underline that E_∞ in (1.1) is the general object of study in the L^∞ -Calculus of Variations, a field initiated by Aronsson in [3]. His pioneering work on the scalar first order case (namely when $N = k = 1$) has been well developed by now, and most challenges have been thoroughly analysed and understood (see [19] for a survey reference). More recently, in [17], the second author approached the vectorial case ($N > 1$), which turned out to be more involved and intriguing than the scalar one. For this reason, a complete theory is still far from reach. The situation is very similar in the higher order case ($k > 1$), recently started in [22, 27]. Without any pretension of being exhaustive, we refer also to [17, 20, 21, 23] for a glimpse of the literature on vectorial first order problems and to [9, 13, 24–26] for higher order ones. Furthermore, the fractional order case ($k \notin \mathbb{N}$) has been recently explored in [10].

It is worth to mention that, given boundary data $u_0 \in W^{k,\infty}(\Omega, \mathbb{R}^N)$, the existence (and/or uniqueness) of absolute minimisers to the Dirichlet problem in $W_{u_0}^{k,\infty}(\Omega, \mathbb{R}^N)$ associated to E_∞ in (1.1) is an open problem already when $k = 1$ and $n, N > 1$ or when $N = 1$ and $n, k > 1$. There are some exceptions in the second order case, in case H has special dependence in the second derivatives of u . We refer to [22, 25, 27] for the scalar second order case and to [9] for the vectorial one (see also [10] for the fractional case).

Now we proceed to specify the assumptions on the function H . By denoting $X = (X_0, X_1, \dots, X_k) \in \mathbb{R}^N \times \mathbb{R}^{Nn} \times \dots \times \mathbb{R}^{Nn^k} = \mathbb{R}^\alpha$, we assume that $H : \Omega \times \mathbb{R}^\alpha \rightarrow \mathbb{R}$ is Carathéodory and radially non-decreasing w.r.t. X , i.e. for almost every $x \in \Omega$ and every $X \in \mathbb{R}^\alpha$ the function

$$t \mapsto H(x, tX) \text{ is non-decreasing in } \mathbb{R}^+.$$

Notice that radial monotonicity implies that, for almost every $x \in \Omega$, $H(x, \cdot)$ has a global minimum at $X = 0$. Without loss of generality we can assume that $H(x, 0) = 0$ a.e. $x \in \Omega$, so that H takes values in $\mathbb{R}^+$ and we can write $E_\infty(u, U) = \|H(\cdot, D^{[k]}u)\|_{L^\infty(U)}$ for every $U \subseteq \Omega$ open.

We point out that any level convex function with global minimum at the origin is necessarily radially non-decreasing, so that our result applies to the class of level convex functions, which is a common assumption for H in L^∞ variational problems (see e.g. [4]). Moreover, one can even apply the result to non-continuous supremands: indeed, for a function $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ lower semicontinuous and non-decreasing, set $G = g \circ H$ and $E'_\infty(u, U) := \text{ess sup}_U G(\cdot, D^{[k]}u)$. It is not difficult to see that E_∞ and E'_∞ share the same minimisers. In addition, if $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ satisfies the maximum principle for E_∞ in Ω , then for every $U \subseteq \Omega$ open

$$E'_\infty(u, U) = g \left(\text{ess sup}_U H(\cdot, D^{[k]}u) \right) = g \left(\text{ess sup}_{\partial U} H(\cdot, D^{[k]}u)^* \right) = \left(\text{ess sup}_{\partial U} G(\cdot, D^{[k]}u)^* \right),$$

i.e. u satisfies the maximum principle for E'_∞ as well.

Now we can state our main result.

Theorem 1.1 (Maximum Principle for Absolute Minimisers). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $H : \Omega \times \mathbb{R}^\alpha \rightarrow \mathbb{R}^+$ be a Carathéodory function with $H(x, 0) = 0$ and $H(x, \cdot)$ radially non-decreasing for a.e. $x \in \Omega$. Assume that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser for the functional E_∞ , defined in (1.1). Then u satisfies the maximum principle for E_∞ in Ω , namely*

$$\text{ess sup}_U H(\cdot, D^{[k]}u) = \sup_{\partial U} H(\cdot, D^{[k]}u)^* \quad \forall U \subseteq \Omega \text{ open.} \quad (1.3)$$

where $H(\cdot, D^{[k]}u)^*$ is the essential limsup of $H(\cdot, D^{[k]}u)$, defined in (1.2).

Before stating our second theorem, we give the definition of absolute minimiser w.r.t. compactly supported variations.

Definition 1.2 (The class $\text{AM}_c(\Omega, \mathbb{R}^N)$). We say that $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$ is an absolute minimiser w.r.t. compactly supported variations, and we write $u \in \text{AM}_c(\Omega, \mathbb{R}^N)$, if

$$E_\infty(u, U) \leq E_\infty(u + \Phi, U) \quad \forall \Phi \in W_c^{k,\infty}(\Omega, \mathbb{R}^N) \quad \forall U \subseteq \Omega \text{ open.}$$

Our second result is the following.

Theorem 1.3. Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $H : \Omega \times \mathbb{R}^\alpha \rightarrow \mathbb{R}^+$ be a Carathéodory function with $H(x, 0) = 0$ and $H(x, \cdot)$ radially non-decreasing for a.e. $x \in \Omega$. Consider the functional E_∞ , defined in (1.1). Then, the following are equivalent for a map $u \in W^{k,\infty}(\Omega, \mathbb{R}^N)$:

- i) u satisfies the maximum principle for E_∞ in Ω ;
- ii) $u \in \text{AM}_c(\Omega, \mathbb{R}^N)$.

The proof of Theorems 1.1 and 1.3 are in Sections, after a short preliminary section.

2 Preparatory tools

2.1 On radial monotonicity

In this first preliminary section, we recall some notions of radial monotonicity for function on $\mathbb{R}^\alpha$, together with some relevant examples.

Definition 2.1. A function $H : \mathbb{R}^\alpha \rightarrow \mathbb{R}$ is said to be

- i) radially non-decreasing if for every $X \in \mathbb{R}^\alpha$ the function $t \mapsto H(tX)$ is non-decreasing in $\mathbb{R}^+$;
- ii) strictly radially increasing if for every $X \in \mathbb{R}^\alpha$ the function $t \mapsto H(tX)$ is strictly increasing in $\mathbb{R}^+$;
- iii) strongly radially increasing if there exists a continuous function $c : [0, 1] \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$, with $c(1, \cdot) = c(\cdot, 0) = 0$ and $c > 0$ in $(0, 1) \times (0, \infty)$, such that

$$H(tX) \leq H(X) - c(t, |X|) \quad \forall t \in [0, 1] \quad \forall X \in \mathbb{R}^\alpha.$$

Notice that $i) \Rightarrow ii) \Rightarrow iii)$, and implications do not invert in general. The above definition is in complete analogy with the notion of level convexity and its stronger versions. In particular, any level convex (resp. strictly level convex) function with global minimum at the origin is radially non-decreasing (resp. strictly radially increasing). Of course, a radially non-decreasing function need not to be level convex, since its level sets need only to be star shaped w.r.t. the origin.

Strongly radially increasing functions can be seen as strict radially increasing functions with a modulus of monotonicity. This class of functions will be particularly important for us, since in the proof of Theorem 1.1 we will use it to approximate radially non-decreasing functions in the uniform convergence.

Example 2.2. Relevant examples of strongly radially increasing functions are:

- Any strongly convex function with global minimum at the origin;
- The cone function associated to a (smooth) open set $S \subset \mathbb{R}^\alpha$ star shaper w.r.t. the origin with $\nu_S(X) \cdot X > 0$ for every $X \in \partial S$, where ν_S is the outward normal vector to ∂S .
- Functions of the form $H_{\lambda,\beta}(X) = H(X) + \lambda|X|^\beta$, for $\lambda, \beta > 0$ and H radially non-decreasing.

2.2 Properties of the essential limsup

In this second preliminary section, we give the definition of essential limsup, together with some properties.

Definition 2.3 (Essential limsup). *Let $U \subseteq \mathbb{R}^n$ be a Borel set and $f \in L^\infty(U)$. We say that the function $f^* \in L^\infty(U)$, defined as*

$$f^*(x) := \lim_{\varepsilon \rightarrow 0} \operatorname{ess\,sup}_{B_\varepsilon(x) \cap U} f \quad \forall x \in U,$$

is the essential limsup of f .

From [18, Proposition 9], we have that $f \leq f^*$ almost everywhere and f^* is upper semicontinuous on U . In general, $f \neq f^*$ on a set of positive measure: for instance, consider $f = \mathbb{1}_{\mathbb{R} \setminus K}$ for a nowhere dense compact set K , with $\mathcal{L}^n(K) > 0$. Moreover, again by [18, Proposition 9], the definition of f^* allows us to give a pointwise meaning to the essential supremum, indeed for any Borel set $U \subseteq \mathbb{R}^n$ we have

$$\sup_U f^* = \operatorname{ess\,sup}_U f. \quad (2.1)$$

For a set $E \subset \mathbb{R}^n$ and $\rho > 0$, we denote by E^ρ the ρ -neighbourhood of E , namely

$$E^\rho := \{x \in \mathbb{R}^n : \operatorname{dist}(x, \partial E) < \rho\}.$$

The following lemma is in order.

Lemma 2.4. *Let $U \subset \mathbb{R}^n$ be an open bounded set. Then for any $K \subseteq \partial U$ compact subset, we have*

$$\sup_K f^* = \lim_{\rho \rightarrow 0} \operatorname{ess\,sup}_{K^\rho \cap U} f.$$

Proof. We start with noticing that for every $x \in K$ there exists $\rho > 0$ such that $B_\rho(x) \cap U \subset K^\rho \cap U$, so that

$$\operatorname{ess\,sup}_{B_\rho(x) \cap U} f \leq \operatorname{ess\,sup}_{K^\rho \cap U} f.$$

By passing to the limit as $\rho \rightarrow 0$ in both sides and taking the supremum on K in the left hand side, we obtain

$$\sup_K f^* \leq \lim_{\rho \rightarrow 0} \operatorname{ess\,sup}_{K^\rho \cap U} f.$$

It remains to prove the opposite inequality. By definition (2.3), for any fixed $x \in K$ and $\sigma > 0$, there exists $\rho_{x,\sigma} > 0$ such that

$$f^*(x) \geq \operatorname{ess\,sup}_{B_{\rho_{x,\sigma}}(x) \cap U} f - \sigma. \quad (2.2)$$

The family of balls $\{B_{\rho_{x,\sigma}}(x); x \in K\}$ is an open covering of K . Thus, we can extract a finite subcovering $\{B_{\rho_{x_i,\sigma}}(x_i); x_i \in K\}_{i=1,\dots,m}$. For seek of brevity, let us denote $B_{i,\sigma} = B_{\rho_{x_i,\sigma}}(x_i)$.

We claim that there exists $\rho_\sigma > 0$ such that $K^{\rho_\sigma} \subset \bigcup_{i=1}^m B_{i,\sigma}$.

By contradiction, suppose that for every $j \in \mathbb{N}$ there exists $y_j \in K^{1/j} \setminus \bigcup_{i=1}^m B_{i,\sigma}$. This is equivalent to say that

$$\forall j \in \mathbb{N} \quad \exists y_j \in \mathbb{R}^n : \begin{cases} y_j \notin B_{i,\sigma} & \forall i = 1, \dots, m, \\ \operatorname{dist}(y_j, K) \leq 1/j. \end{cases} \quad (2.3)$$

Let $\bar{y}_j \in K$ be a point realising $d(y_j, K)$. Since K is compact, there exists a subsequence $(\bar{y}_{j_k}) \subset (\bar{y}_j)$ such that $\bar{y}_{j_k} \rightarrow \bar{x}$ as $k \rightarrow \infty$, for some $\bar{x} \in K$. Moreover, since $\text{dist}(y_{j_k}, \bar{y}_{j_k}) \leq 1/j_k$, we have also that $y_{j_k} \rightarrow \bar{x}$ as $k \rightarrow \infty$. But being $\{B_{i,\sigma}\}_{i=1,\dots,m}$ a covering of K , we must have that eventually $y_{j_k} \in B_{i,\sigma}$ for some $i \in \{1, \dots, m\}$, which contradicts (2.3). Now, define $U_\sigma = \bigcup_{i=1}^m (B_{i,\sigma} \cap U)$ and notice that $K^\rho \cap U \subset U_\sigma$. Thus, using also (2.2), we have

$$\begin{aligned} \lim_{\rho \rightarrow 0} \text{ess sup}_{K^\rho \cap U} f &\leq \text{ess sup}_{K^\rho \sigma \cap U} f \\ &\leq \text{ess sup}_{U_\sigma} f \\ &\leq \max_{i=1,\dots,m} \text{ess sup}_{B_{i,\sigma} \cap U} f \\ &\leq \max_{i=1,\dots,m} f^*(x_i) + \sigma \\ &\leq \sup_K f^* + \sigma. \end{aligned}$$

By letting $\sigma \rightarrow 0$, we conclude the proof. $\square$

3 Proof of the main results

We divide the proof of Theorem ?? into three parts, concerning existence, PDE derivation and uniqueness respectively. In the first two, our arguments follow similar lines to those in [22], appropriately extended to the vectorial setting up to some technical issues, while in the last part our arguments are significantly different.

3.1 Existence

Let $p \geq 1$ and recall that $u_0 \in \mathcal{W}^{2,\infty}(\Omega, \mathbb{R}^N)$. We begin by considering the functional

$$E_p(u) := \left(\int_{\Omega} F(x, Lu(x))^p dx \right)^{\frac{1}{p}}, \quad u \in W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N).$$

Assume that $E_p(u) < \infty$ for some $u \in W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N)$. By making use of (??) and Corollary ??, we have $Lu \in L^{2p}(\Omega, \mathbb{R}^N)$ and

$$\|u\|_{2,2p} \leq C (\|Lu\|_{2p} + \|u_0\|_{2,2p}) \leq C \left(c^{\frac{1}{2}} |\Omega|^{\frac{1}{2p}} E_p(u)^{\frac{1}{2}} + \|u_0\|_{2,2p} \right)$$

for a constant $C = C(\Omega, p, \lambda, A)$. Thus, we deduce that E_p is coercive in $W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N)$. Moreover, since F is convex with respect to ξ , by Direct Methods we obtain the existence of a minimiser $u_p \in W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N)$ of E_p .

Fix $k \in \mathbb{N}$. Extend E_p and E_∞ to $W_{u_0}^{2,k}(\Omega, \mathbb{R}^N)$ as

$$E_p(u) = \begin{cases} \left(\int_{\Omega} F(x, Lu(x))^p dx \right)^{\frac{1}{p}}, & u \in W_{u_0}^{2,2p}, \\ +\infty & u \in W_{u_0}^{2,k} \setminus W_{u_0}^{2,2p}, \end{cases} \quad E_\infty(u) = \begin{cases} \|F(\cdot, Lu)\|_\infty, & u \in \mathcal{W}_{u_0}^{2,\infty}, \\ +\infty & u \in W_{u_0}^{2,k} \setminus \mathcal{W}_{u_0}^{2,\infty}. \end{cases}$$

Let us show that $(E_p)_p$ is equi-coercive in $W_{u_0}^{2,k}$ with respect to p . Assume $E_p(u) \leq M$ for all $p \in [k/2, \infty)$. By (??) and the Hölder inequality, we have

$$c^{-\frac{1}{2}} |\Omega|^{-\frac{1}{2k}} \|Lu\|_k \leq E_k(u)^{\frac{1}{2}} \leq E_{2p}(u)^{\frac{1}{2}} \leq M^{\frac{1}{2}}. \quad (3.1)$$

Hence, by Corollary ??

$$\|u\|_{2,k} \leq C (\|Lu\|_k + \|u_0\|_{2,k}) \leq C \left(c^{\frac{1}{2}} |\Omega|^{\frac{1}{k}} M^{\frac{1}{2}} + \|u_0\|_{2,k} \right),$$

for a constant $C = C(\Omega, k, \lambda, A)$. This gives the equi-coercivity of $(E_p)_p$ in $W_{u_0}^{2,k}$.

Moreover, $(E_p)_p$ is a monotone increasing sequence of weakly lower semicontinuous functionals on $W_{u_0}^{2,k}$, thus its Γ -limit is given by (see [6, Remark 2.12])

$$\Gamma - \lim_p E_p = \lim_p E_p = E_\infty.$$

By the Fundamental Theorem of Gamma Convergence [6, Thm. 2.10], if $(u_p)_p$ is a sequence of minimisers of E_p , up to passing to a subsequence, we have $u_p \rightharpoonup u_\infty$ weakly $W_{u_0}^{2,k}$ where u_∞ is a minimiser of E_∞ on $W_{u_0}^{2,k}$. By definition of E_∞ , we necessarily have $u_\infty \in \mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$. Moreover, by the same Theorem

$$E_p(u_p) \rightarrow E_\infty(u_\infty) \quad \text{as } p \rightarrow \infty. \quad (3.2)$$

3.2 PDE derivation

Now we work towards the proof of (??). Let us begin by writing down the system of Euler-Lagrange equations satisfied by u_p , using the variational nature of the operator L and the symmetry of A :

$$Lf_p = 0, \quad \text{in } \mathcal{D}'(\Omega, \mathbb{R}^N), \quad (3.3)$$

where, for $e_p := E_p(u_p)$, $f_p : \Omega \rightarrow \mathbb{R}^N$ is defined by

$$f_p := e_p^{1-p} F(\cdot, Lu_p)^{p-1} F_\xi(\cdot, Lu_p). \quad (3.4)$$

Since the only interesting case is $e_\infty > 0$, we may assume that $e_p \geq \alpha > 0$ (for p large enough). In fact, if $e_\infty = 0$, then $F(\cdot, Lu_\infty) = 0$ and system (??) reduces to $Lu_\infty = 0$. For this reason, f_p is a well defined measurable map.

Let us prove that that $(f_p)_p$ is bounded in $L^1(\Omega, \mathbb{R}^N)$:

$$\begin{aligned} \int_\Omega |f_p(x)| dx &= e_p^{1-p} \int_\Omega F(x, Lu_p(x))^{p-1} |F_\xi(x, Lu_p(x))| dx \\ &\leq c e_p^{1-p} \int_\Omega F(x, Lu_p(x))^{p-1} |Lu_p(x)| dx \\ &\leq c e_p^{1-p} \left(\int_\Omega F(x, Lu_p(x))^p dx \right)^{\frac{p-1}{p}} \left(\int_\Omega |Lu_p(x)|^p dx \right)^{\frac{1}{p}} \\ &= c \left(\int_\Omega |Lu_p(x)|^p dx \right)^{\frac{1}{p}} \\ &\leq c \left(\int_\Omega [cF(x, Lu_p(x))]^{\frac{p}{2}} dx \right)^{\frac{1}{p}} \\ &\leq c c^{\frac{1}{2}} \left(\int_\Omega F(x, Lu_p(x))^p dx \right)^{\frac{1}{2p}} = c^{\frac{3}{2}} e_p^{\frac{1}{2}} \leq c^{\frac{3}{2}} e_\infty^{\frac{1}{2}}. \end{aligned} \quad (3.5)$$

In particular, thanks to Proposition ??, $f_p \in W_{loc}^{1,2}(\Omega, \mathbb{R}^N) \cap L^1(\Omega, \mathbb{R}^N)$ is a weak solution of (3.3). Thus, by Corollary ??, for every $q < \infty$,

$$\|f_p\|_{2,q,U} \leq C \|f_p\|_1 \leq C, \quad \forall U \Subset \Omega,$$

for some constant $C = C(q, U, \Omega, c, e_\infty) > 0$. Hence, by Morrey's theorem, we deduce that $(f_p)_p$ is bounded in the local topology of C^1 , and so there exists f_∞ and a subsequence $(f_{p_\ell})_\ell$ such that $f_{p_\ell} \rightarrow f_\infty$ locally uniformly as $\ell \rightarrow \infty$. In particular, $Lf_\infty = 0$ in the sense of distributions and $f_\infty \in L^1(\Omega, \mathbb{R}^N)$ as well.

The next lemma is central in our argument.

Lemma 3.1. (Non-triviality of f_∞) *The map f_∞ constructed above is such that $f_\infty \not\equiv 0$.*

Proof. First we show that $f_p \xrightarrow{*} f_\infty$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$, as $p \rightarrow \infty$ along a subsequence.

By approximating with C_c^∞ -maps, thanks to the regularity assumptions on A , for every $p \in (1, \infty)$ we have

$$Lf_p = Lf_\infty = 0 \quad \text{in } \left(W_0^{2,\infty}(\Omega, \mathbb{R}^N)\right)^*. \quad (3.6)$$

From the boundedness in L^1 , there exists a measure $\mu \in \mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$ such that $f_p \mathcal{L}^n \llcorner \Omega \xrightarrow{*} \mu$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$. Since $f_p \rightarrow f_\infty$ in $W_{loc}^{2,q}(\Omega, \mathbb{R}^N)$, we necessarily have $\mu \llcorner \Omega = f_\infty \mathcal{L}^n \llcorner \Omega$. So we can write

$$\mu = f_\infty \mathcal{L}^n \llcorner \Omega + \mu \llcorner \partial\Omega.$$

Now we claim that $\mu \llcorner \partial\Omega = 0$. Let us denote $d := \text{dist}(\cdot, \partial\Omega)$ and $\Omega_r := \{x \in \Omega : d(x) \leq r\}$. Then, by the regularity of $\partial\Omega$, we have $d \in C^2(\overline{\Omega}_r)$ for some $r > 0$ sufficiently small. Extend it to a function $d \in C^2(\overline{\Omega})$ and fix a map $g \in C^2(\overline{\Omega}, \mathbb{R}^N)$. Set $\psi := d^2 g \in C^2(\overline{\Omega}, \mathbb{R}^N) \cap W_0^{2,\infty}(\Omega, \mathbb{R}^N)$ and test (3.6) with ψ :

$$0 = \int_{\Omega} (f_p - f_\infty) \cdot L\psi = \int_{\Omega} (f_p - f_\infty) \cdot [(\text{div} A) D\psi + A D^2 \psi]. \quad (3.7)$$

By a computation,

$$\begin{aligned} D\psi &= 2dg \otimes Dd + d^2 Dg, \\ D^2 \psi &= 2g \otimes Dd \otimes Dd + 2dg \otimes D^2 d + 4d Dg \otimes Dd + d^2 D^2 g. \end{aligned}$$

Thus,

$$D^2 \psi \llcorner \partial\Omega = 2(g \otimes Dd \otimes Dd) \llcorner \partial\Omega = 2g \otimes \nu \otimes \nu \llcorner \partial\Omega,$$

where ν is the outwards unit normal to the boundary. Since $(\text{div} A) D\psi \in C_0(\Omega, \mathbb{R}^N)$ and $(f_p - f_\infty) \mathcal{L}^n \llcorner \Omega \xrightarrow{*} \mu \llcorner \partial\Omega$ in $\mathcal{M}(\overline{\Omega}, \mathbb{R}^N)$, we can pass to the limit in (3.7) to obtain

$$0 = 2 \int_{\partial\Omega} A : g \otimes \nu \otimes \nu \cdot d\mu.$$

By the ellipticity of A , the matrix $M := A : \nu \otimes \nu$ is positive definite, so we conclude that

$$\int_{\partial\Omega} Mg \cdot d\mu = 0, \quad \forall g \in C^2(\overline{\Omega}, \mathbb{R}^N),$$

which gives $\mu \llcorner \partial\Omega = 0$.

By Lemma ??, we estimate

$$\begin{aligned} \int_{\Omega} f_p \cdot Lu_p &= e_p^{1-p} \int_{\Omega} F(x, Lu_p(x))^{p-1} F_\xi(x, Lu_p(x)) \cdot Lu_p(x) dx \\ &\geq e_p^{1-p} \int_{\Omega} F(x, Lu_p(x))^p dx \\ &= e_p^{1-p} e_p^p |\Omega| = |\Omega| e_p \geq |\Omega| \alpha, \end{aligned}$$

where in the last step, we recall we are under the assumption $e_\infty > 0$.

Thanks to Whitney's extension theorem, there exists $u_* \in C^2(\mathbb{R}^n, \mathbb{R}^N)$ such that $u_* = u_0$ and $Du_* = Du_0$ on $\partial\Omega$. Now we observe that (3.3) clearly holds also in $(W_0^{2,2p}(\Omega, \mathbb{R}^N))^*$, therefore by testing with $u_p - u_* \in W_0^{2,2p}(\Omega, \mathbb{R}^N)$, we obtain

$$|\Omega|\alpha \leq \int_{\Omega} f_p \cdot Lu_p = \int_{\Omega} f_p \cdot Lu_*.$$

So, we can pass to the limit on the right hand side, observing that $Lu_* \in C(\bar{\Omega}, \mathbb{R}^N)$, and conclude

$$\int_{\Omega} f_\infty \cdot Lu_* \geq |\Omega|\alpha.$$

In particular, this ensures that $f_\infty \not\equiv 0$. □

Thanks to Lemma 3.1, we are now able to derive the PDE satisfied by u_∞ .

Set $\Gamma_\infty := f_\infty^{-1}(0) \subsetneq \Omega$. From (3.4), we obtain the equation

$$|f_p(x)|^{\frac{2}{2p-1}} \frac{f_p(x)}{|f_p(x)|} = e_p^{-1+\frac{1}{2p-1}} F(x, Lu_p(x)) \left(\frac{|F_\xi(x, Lu_p(x))|}{\sqrt{F}(x, Lu_p(x))} \right)^{\frac{2}{2p-1}} \frac{F_\xi(x, Lu_p(x))}{|F_\xi(x, Lu_p(x))|}, \quad \text{a.e. } x \in \Omega, \quad (3.8)$$

which we rewrite as

$$e_p^{1-\frac{1}{2p-1}} \left(\frac{|F_\xi(x, Lu_p(x))|}{\sqrt{F}(x, Lu_p(x))} \right)^{-\frac{2}{2p-1}} |f_p(x)|^{\frac{2}{2p-1}} \frac{f_p(x)}{|f_p(x)|} = \Phi(x, Lu_p(x)), \quad \text{a.e. } x \in \Omega, \quad (3.9)$$

where we have symbolised

$$\Phi(x, \xi) := F(x, \xi) \frac{F_\xi(x, \xi)}{|F_\xi(x, \xi)|}, \quad \text{a.e. } x \in \Omega, \quad \forall \xi \in \mathbb{R}^N.$$

Notice that equations (3.8) and (3.9) are well defined. Indeed, from (??), (??), and Lemma ??, it is easy to see that $|F_\xi|/\sqrt{F}$ is bounded from above and below. Moreover, by the assumptions on F , we have $f_p = 0$ on a set of positive measure $E \subset \Omega$ if and only if $Lu_p = 0$ on the same set. So, points in E are not singular for both sides of (3.8) and (3.9), naturally vanishing on E .

Now we recall that by Lemma ??, for almost every $x \in \Omega$, the map $\Phi_x := \Phi(x, \cdot) : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a homeomorphism. So, applying the inverse map Φ_x^{-1} , we rewrite the previous equation as

$$Lu_p(x) = \Phi_x^{-1} \left(e_p^{1-\frac{1}{2p-1}} \left(\frac{|F_\xi(x, Lu_p(x))|}{\sqrt{F}(x, Lu_p(x))} \right)^{-\frac{2}{2p-1}} |f_p(x)|^{\frac{2}{2p-1}} \frac{f_p(x)}{|f_p(x)|} \right), \quad \text{a.e. } x \in \Omega.$$

Recall that $f_{p_\ell} \rightarrow f_\infty$ uniformly on every compact $K \subset \Omega \setminus \Gamma_\infty$ and, by (3.2), $e_{p_\ell} \rightarrow e_\infty$ as $\ell \rightarrow \infty$. So we can pass to the limit locally uniformly in the right hand side along the subsequence $(p_\ell)_\ell$. On the other hand, since we already know that $Lu_p \rightharpoonup Lu_\infty$ weakly in L^q for every $q \geq 1$, we obtain

$$Lu_\infty(x) = \Phi_x^{-1} \left(e_\infty \frac{f_\infty(x)}{|f_\infty(x)|} \right), \quad \text{a.e. } x \in K.$$

By inverting again, we get

$$\Phi(x, Lu_\infty) = F(x, Lu_\infty) \frac{F_\xi(x, Lu_\infty)}{|F_\xi(x, Lu_\infty)|} = e_\infty \frac{f_\infty(x)}{|f_\infty(x)|}, \quad \text{a.e. } x \in K.$$

Since $Lf_\infty = 0$, the unique continuation property of L yields $|\Gamma_\infty| = 0$. As a consequence, we obtain (??).

3.3 Uniqueness

In the previous sections, we proved the existence of a minimiser $u_\infty \in \mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$ of E_∞ and that it satisfies the PDE system (??). As a first step to provide uniqueness of minimisers of E_∞ in $\mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$, we show that every minimiser of E_∞ satisfies a PDE system of the same structure as (??). Some of the arguments are similar to Section 3.1, but we will present the complete proof for sake of clarity. More precisely, we start with proving the following result.

Lemma 3.2. (Necessity of the PDE system) *Let F, A , and u_0 be as in Theorem ???. Suppose that $u \in \mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$ is a minimiser of E_∞ . Then there exists $f \in L^1(\Omega, \mathbb{R}^N)$ such that $f \in W_{loc}^{2,q}(\Omega, \mathbb{R}^N)$ for all $q < \infty$ and*

$$\begin{cases} F(x, Lu(x)) \frac{F_\xi(x, Lu(x))}{|F_\xi(x, Lu(x))|} = e_\infty \frac{f(x)}{|f(x)|}, & \text{a.e. } x \in \Omega, \\ Lf(x) = 0, & \text{a.e. } x \in \Omega. \end{cases} \quad (3.10)$$

Proof. Let $u \in \mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$ be a minimiser of E_∞ . For any $p \geq 1$, we consider the auxiliary functional

$$A_p(v) := E_p(v) + \frac{1}{2} \int_{\Omega} |v - u|^2, \quad v \in W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N).$$

Using the Direct Method, it is not difficult to see that A_p attains a minimum point. Let $v_p \in W_{u_0}^{2,2p}(\Omega, \mathbb{R}^N)$ be a minimiser of A_p , then, recalling that u_p is a minimiser of E_p in the same space, we have

$$e_p = E_p(u_p) \leq E_p(v_p) \leq A_p(v_p) \leq A_p(u) = E_p(u) \leq E_\infty(u) = e_\infty.$$

Since $e_p \rightarrow e_\infty$, we have $A_p(v_p) \rightarrow e_\infty$ as $p \rightarrow \infty$. Now, if $(v_p)_p$ is a sequence of minimisers of A_p , by Calderon-Zygmund estimates, we can prove as in Section 3.1 that $(v_p)_p$ bounded in $W_{u_0}^{2,q}(\Omega, \mathbb{R}^N)$ for any fixed $q \geq 1$. So, $v_{p_\ell} \rightharpoonup v_\infty$ weakly $W_{u_0}^{2,q}$ for a subsequence $(p_\ell)_\ell$ and some $v_\infty \in W_{u_0}^{2,q}(\Omega, \mathbb{R}^N)$. We claim that $v_\infty = u$. Indeed, passing to the limit as $\ell \rightarrow \infty$ in the following inequality

$$e_{p_\ell} + \frac{1}{2} \int_{\Omega} |v_{p_\ell} - u|^2 \leq A_{p_\ell}(v_{p_\ell}),$$

we get

$$e_\infty + \frac{1}{2} \int_{\Omega} |v_\infty - u|^2 \leq e_\infty,$$

which necessarily implies $v_\infty = u$.

Now, for simplicity let us denote $v_p = v_{p_\ell}$ and write down the Euler-Lagrange equation satisfied by v_p :

$$Lg_p + v_p - u = 0 \quad \text{in } \mathcal{D}'(\Omega, \mathbb{R}^N), \quad (3.11)$$

where, for $a_p := E_p(v_p)$, g_p is defined by $g_p := a_p^{1-p} F(\cdot, Lv_p)^{p-1} F_\xi(\cdot, Lv_p)$. We can prove that g_p is well defined and bounded in L^1 as showed for f_p in Section 3.1. In particular, thanks to Proposition ??, $g_p \in W_{loc}^{1,2}(\Omega, \mathbb{R}^N) \cap L^1(\Omega, \mathbb{R}^N)$ is a weak solution of (3.11). Thus, by Corollary ??, for every $q < \infty$,

$$\|g_p\|_{2,q,U} \leq C (\|g_p\|_1 + \|v_p - u\|_q) \leq C, \quad \forall U \Subset \Omega,$$

for some constant $C = C(q, U, \Omega, c, e_\infty) > 0$. By Sobolev embeddings, up to passing to a (not relabeled) subsequence, $g_p \rightarrow g_\infty$ locally uniformly as $p \rightarrow \infty$, for some $g_\infty \in L^1(\Omega, \mathbb{R}^N) \cap W_{loc}^{2,q}(\Omega, \mathbb{R}^N)$ such that $Lg_\infty = 0$. We can prove that $g_\infty \not\equiv 0$ as in Lemma 3.1, since the same argument holds even if $Lg_p \rightharpoonup 0$ in the sense of distributions only. Finally, since $a_p \rightarrow e_\infty$ as $p \rightarrow \infty$, we can pass to the limit in the expression defining g_p exactly as we did for f_p , obtaining (3.10) with $f = g_\infty$. $\square$

We conclude this section with the proof of uniqueness of the solution to the minimum problem associated to E_∞ . Let u_1 and u_2 be two minimisers in $\mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$, then, by (3.10), we have

$$F(\cdot, Lu_1) = F(\cdot, Lu_2) = e_\infty, \quad \text{a.e. in } \Omega.$$

By convexity of $F(x, \cdot)$ and linearity of L , for a.e. $x \in \Omega$ we have

$$\begin{aligned} F\left(x, L\left(\frac{u_1(x) + u_2(x)}{2}\right)\right) &= F\left(x, \frac{Lu_1(x)}{2} + \frac{Lu_2(x)}{2}\right) \\ &\leq \frac{1}{2}F(x, Lu_1(x)) + \frac{1}{2}F(x, Lu_2(x)) \\ &= e_\infty. \end{aligned} \tag{3.12}$$

The average $\frac{u_1+u_2}{2}$ still belongs to $\mathcal{W}_{u_0}^{2,\infty}(\Omega, \mathbb{R}^N)$, then necessarily

$$F\left(\cdot, L\left(\frac{u_1 + u_2}{2}\right)\right) = e_\infty, \quad \text{a.e. in } \Omega.$$

But $F(x, \cdot)$ is strictly convex, so the only possibility to have equality in (3.12) is that $Lu_1 = Lu_2$, giving $u_1 = u_2$ by uniqueness results for elliptic systems with Dirichlet boundary conditions (see [14, Theorems 3.39 and 3.46]).

4 On the unique continuation property

In this last section, we identify classes of operators satisfying the unique continuation property (see Definition ??). First, it is worth mentioning that a slightly different notion that often appears in the literature is the *strong unique continuation property*, namely the only solution to $Lu = 0$ with a zero of infinite order is the trivial one. We recall that $x_0 \in \Omega$ is a zero of infinite order for u if

$$\int_{B_r(x_0)} |u|^2 dx = O(r^k) \quad \text{as } r \rightarrow 0 \quad \forall k \in \mathbb{N}.$$

There is also the definition of *weak unique continuation property* (the only solution to $Lu = 0$ vanishing on an open set is the trivial one), but we are not interested in that one. For a proof of the fact that the strong definition implies the one in Definition ??, we refer to [15]. Their proof actually applies to elliptic equations, but it can be identically readapted for systems, since it is basically a consequence of Caccioppoli inequalities [14, Sec. 4.2]. However, differently from the scalar case, where the zero nodal set is even $(n-1)$ -dimensional [16], we have no theory available that guarantees sufficient conditions for (strong) unique continuation property for elliptic systems. For this reason, we restrict ourself to list some examples of interest, although one of them is deeply connected to the scalar case. In what follows, we assume that $u \in L^1(\Omega, \mathbb{R}^N)$ is a distributional solution of $Lu = \operatorname{div}(ADu) = 0$ with A satisfying either (H1) or (H2). Notice that by Proposition ??, u is actually a weak solution. Without loss of generality, we assume that u is not constant. We consider the following cases:

- A has analytic coefficients.

In this case, it is well known that solutions to $Lu = 0$ are analytic [37]. In particular, if u is not constant, its nodal sets are negligible.

- A has the form

$$A = \sum_{h=1}^N e_h \otimes e_h \otimes B_h$$

with $\{e_h\}_{h=1,\dots,N}$ orthonormal base of $\mathbb{R}^N$ and $B_h \in C^1(\overline{\Omega}, \mathbb{R}^{n \times n})$ elliptic for every $h = 1, \dots, N$.

We have

$$(Lu)_i = D_\alpha \left(\left[\sum_{h=1}^N e_{h_i} e_{h_j} B_h^{\alpha\beta} \right] D_\beta u^j \right) = 0 \quad \forall i = 1, \dots, N.$$

Fix $k \in \{1, \dots, N\}$ and take the scalar product of Lu with e_k :

$$\begin{aligned} 0 &= e_k^i (Lu)_i \\ &= D_\alpha \left(\left[\sum_{h=1}^N e_k^i e_{h_i} e_{h_j} B_h^{\alpha\beta} \right] D_\beta u^j \right) \\ &= D_\alpha \left(e_{k_j} B_k^{\alpha\beta} D_\beta u^j \right) \\ &= D_\alpha \left(B_k^{\alpha\beta} D_\beta (e_{k_j} u^j) \right) \end{aligned}$$

Set $u^{(k)} = e_k \cdot u$, namely the k -th component of u w.r.t. the base $\{e_h\}$. Then, for each $k = 1, \dots, N$, $u^{(k)}$ solves an elliptic equation in divergence form. Since we assumed that u is not constant, there is at least one k such that $u^{(k)}$ is not constant. So, by applying the scalar theory [16] to $u^{(k)}$, since $\{|u| = 0\} \subset \{u^{(k)} = 0\}$, the unique continuation must hold.

ACKNOWLEDGEMENTS: S.C and R.M. acknowledge partial financial support through the EPSRC grant EP/X017206/1. S.C and N.K. acknowledge partial financial support through the EPSRC grant EP/X017109/1. S.C. is a member of Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni (GNAMPA) of the Istituto Nazionale di Alta Matematica (INdAM) of Italy.

DATA AVAILABILITY: Our manuscript has no associated data.

CONFLICT OF INTEREST: The authors have no Conflict of interest to declare for this article.

References

- [1] L. Ambrosio, A. Carlotto, A. Massaccesi, *Lectures on Elliptic Partial Differential Equations*, Publications of the Scuola Normale Superiore (2019).
- [2] N. Ansini, F. Prinari, *On the lower semicontinuity of supremal functional under differential constraints*, ESAIM - Control, Opt. and Calc. Var., **21** (4), 1053–1075 (2015).
- [3] G. Aronsson, *Minimization problems for the functional $\sup_x F(x, f(x), f'(x))$ I, II, III*, Arkiv fur Mat. 33 - 53 (1965); 409 - 431 (1966); 509 - 512 (1969).
- [4] E.N. Barron, R.R. Jensen, C.Y. Wang, *The Euler equation and absolute minimizers of L^∞ functionals*, Arch. Rational Mech. Anal. **157**, 255-283 (2001).
- [5] V. I. Bogachev, N. V. Krylov, M. Röckner, *Elliptic and parabolic equations for measures*, Uspekhi Mat. Nauk **64**, 5–116 (2009), translation in Russian Math. Surveys **64**, 973–1078 (2009).

- [6] A. Braides, *A handbook of Γ -convergence*, Handbook of Differential Equations: Stationary Partial Differential Equations, North-Holland, **3**, 101-213 (2006).
- [7] C. Brizzi, L. De Pascale, *A property of Absolute Minimizers in L^∞ Calculus of Variations and of solutions of the Aronsson-Euler equation*, Advances in Differential Equations 28 (3/4), 287-310 (2023).
- [8] L. Bungert, Y. Korolev, *Eigenvalue problems in L^∞ : optimality conditions, duality, and relations with optimal transport*, Communications of the American Mathematical Society 2:345 (2022).
- [9] S. Carano, N. Katzourakis, R. Moser, *Existence, uniqueness and characterisation of vector-valued absolute minimisers for a second order L^∞ -variational problem*, Preprint arXiv <https://arxiv.org/abs/2504.04181> (2025).
- [10] S. Carano, R. Moser, *An L^∞ -variational problem involving the Fractional Laplacian*, Preprint arXiv <https://arxiv.org/abs/2510.14476> (2025).
- [11] B. Dacorogna, *Direct Methods in the Calculus of Variations*, 2nd ed., Appl. Math. Sci. 78, Springer, New York, 2008.
- [12] J. M. Danskin, *The theory of max-min with applications*, J. SIAM Appl. Math. **14**(4), 641-665 (1966).
- [13] B. Dutton, N. Katzourakis, *On Second-Order L^∞ Variational Problems with Lower-Order Terms*, ArXiv Preprint <https://arxiv.org/abs/2412.11701> (2024).
- [14] M. Giaquinta, L. Martinazzi, *An Introduction to the Regularity Theory for Elliptic Systems, Harmonic Maps and Minimal Graphs*, Publications of the Scuola Normale Superiore (2018).
- [15] D. G. de Figueiredo, J. P. Gossez, *Strict Monotonicity of Eigenvalues and Unique Continuation*, Communications in Partial Differential Equations **17**, 339–346 (1992).
- [16] R. Hardt, L. Simon, *Nodal sets for solutions of elliptic equations*, J. Differential Geom. **30** 505 - 522 (1989).
- [17] N. Katzourakis, *L^∞ Variational Problems for maps and the Aronsson PDE System*, J. Differential Equations **253**(7), 2123 - 2139 (2012).
- [18] N. Katzourakis, *An L^∞ regularisation strategy to the inverse source identification problem for elliptic equations*, SIAM.....
- [19] N. Katzourakis, *An Introduction to Viscosity Solutions for Fully Nonlinear PDE with Applications to Calculus of Variations in L^∞* , Springer Briefs in Mathematics, DOI 10.1007/978-3-319-12829-0 (2015).
- [20] N. Katzourakis, *∞ -minimal submanifolds*, Proc. Amer. Math. Soc. **142**, 2797-2811 (2014).
- [21] N. Katzourakis, *On the structure of ∞ -harmonic maps*, Comm. Partial Differential Equations **39**, 2091–2124 (2014).
- [22] N. Katzourakis, R. Moser, *Existence, uniqueness and structure of second order absolute minimisers*, ARMA **231**(3), 1615-1634 (2019).

- [23] N. Katzourakis, R. Moser, *Minimisers of supremal functionals and mass-minimising 1-currents*, Calc. Var. **64**(26) (2025).
- [24] N. Katzourakis, R. Moser, *Variational problems in L^∞ involving semilinear second order differential operators*, ESAIM: COCV, **29**(76) (2023).
- [25] N. Katzourakis, R. Moser, *Existence, uniqueness and characterisation of local minimisers in higher order Calculus of Variations in L^∞* , ArXiv Preprint, <https://arxiv.org/abs/2403.12625> (2024).
- [26] N. Katzourakis, T. Pryer, *On the numerical approximation of p -Biharmonic and ∞ -Biharmonic functions*, Numerical Methods for Partial Differential Equations, **35**(1), 155-180 (2018).
- [27] N. Katzourakis, T. Pryer, *2nd order L^∞ Variational Problems and the ∞ -Polylaplacian*, Advances in Calculus of Variations **13**, 115-140 (2020).
- [28] C. Kreisbeck, E. Zappale, *Lower semicontinuity and relaxation of nonlocal L^∞ -functionals*, Calculus of Variations and PDE, **59** (138), 1–36 (2020).
- [29] M. Manetti, *Topology*, Springer Cham, DOI 10.1007/978-3-319-16958-3 (2015).
- [30] Q. Miao, C. Wang, Y. Zhou, *Uniqueness of Absolute Minimizers for L^∞ -Functionals Involving Hamiltonians $H(x, p)$* , Archive for Rational Mechanics and Analysis **223** (1), 141–198 (2017).
- [31] R. Moser, *Structure and classification results for the ∞ -elastica problem*, American Journal of Mathematics **144**(5), 1299-1329 (2022).
- [32] R. Moser, H. Schwetlick, *Minimizers of a weighted maximum of the Gauss curvature*, Annals of Global Analysis and Geometry **41**, 199-207 (2012).
- [33] F. Peng, C. Wang, Y. Zhou, *Regularity of absolute minimizers for continuous convex Hamiltonians*, Journal of Differential Equations, Volume 274 (2021), 1115-1164.
- [34] F. Prinari, E. Zappale, *A Relaxation Result in the Vectorial Setting and Power Law Approximation for Supremal Functionals*, J Optim. Theory Appl., **186**, 412–452 (2020).
- [35] A. M. Ribeiro, E. Zappale, *Existence of minimisers for nonlevel convex functionals*, SIAM J. Control Opt., **52** (5), 3341–3370 (2014).
- [36] A. M. Ribeiro, E. Zappale, *Revisited convexity notions for L^∞ variational problems*, Revista Matemática Complutense, published online, <https://doi.org/10.1007/s13163-024-00499-0> (2024).
- [37] O. A. Oleĭnik, E. V. Radkevič, *On the analyticity of solutions of linear partial differential equations*, Math. USSR Sb. **19**(581) (1973).
- [38] H. Whitney, *Analytic Extensions of Differentiable Functions Defined in Closed Sets*, Transactions of the American Mathematical Society **36**(1), 63-89 (1934).

SIMONE CARANO
 Department of Mathematical Sciences, University of Bath
 Bath BA2 7AY, UK

E-mail: sc3705@bath.ac.uk

NIKOS KATZOURAKIS

Department of Mathematics and Statistics, University of Reading
Whiteknights Campus, Pepper Lane, Reading RG6 6AX, UK
E-mail: n.katzourakis@reading.ac.uk

ROGER MOSER

Department of Mathematical Sciences, University of Bath
Bath BA2 7AY, UK
E-mail: r.moser@bath.ac.uk